



# CloudMinds Robotics

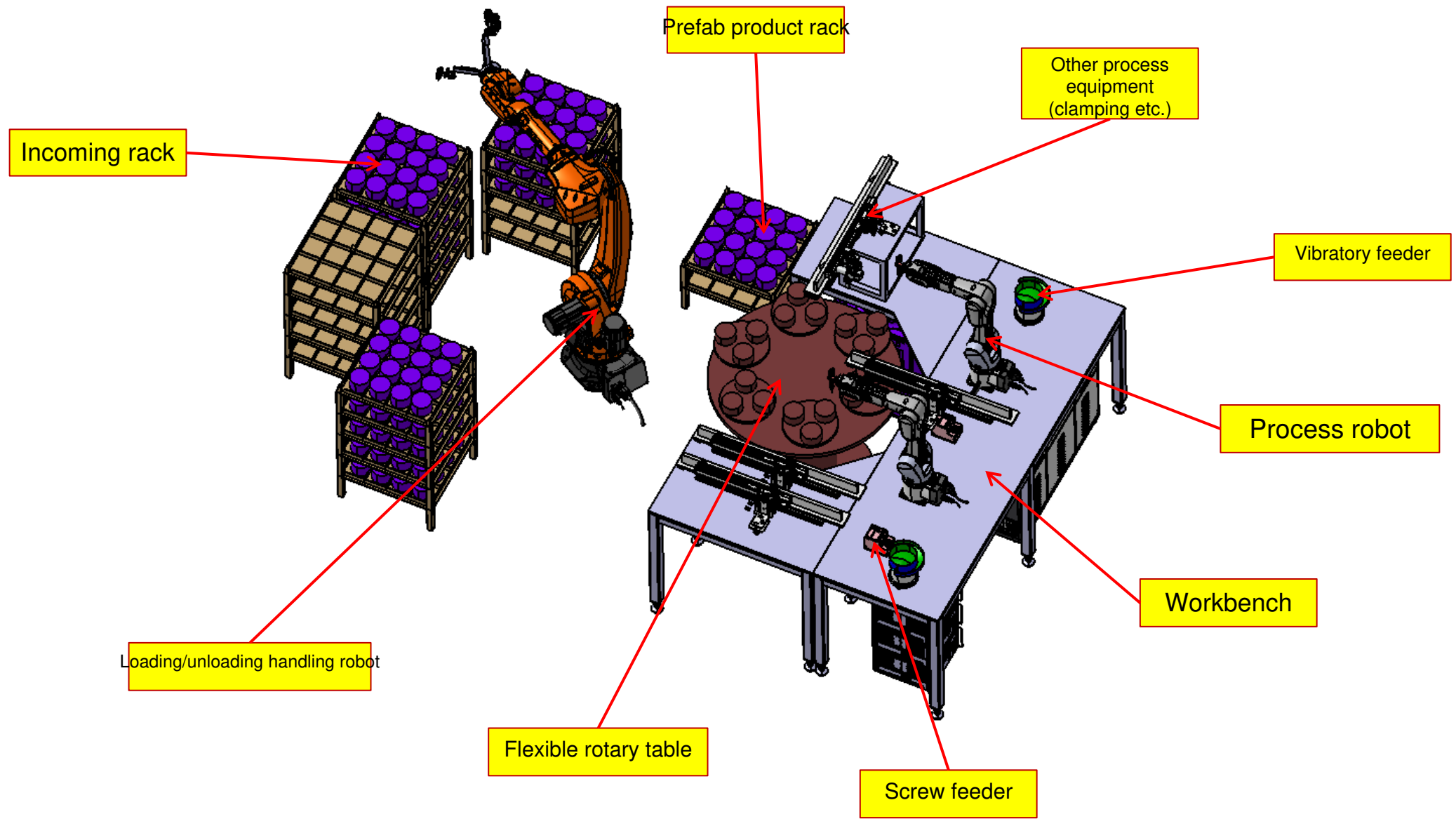
## Standardized Workstation Solution





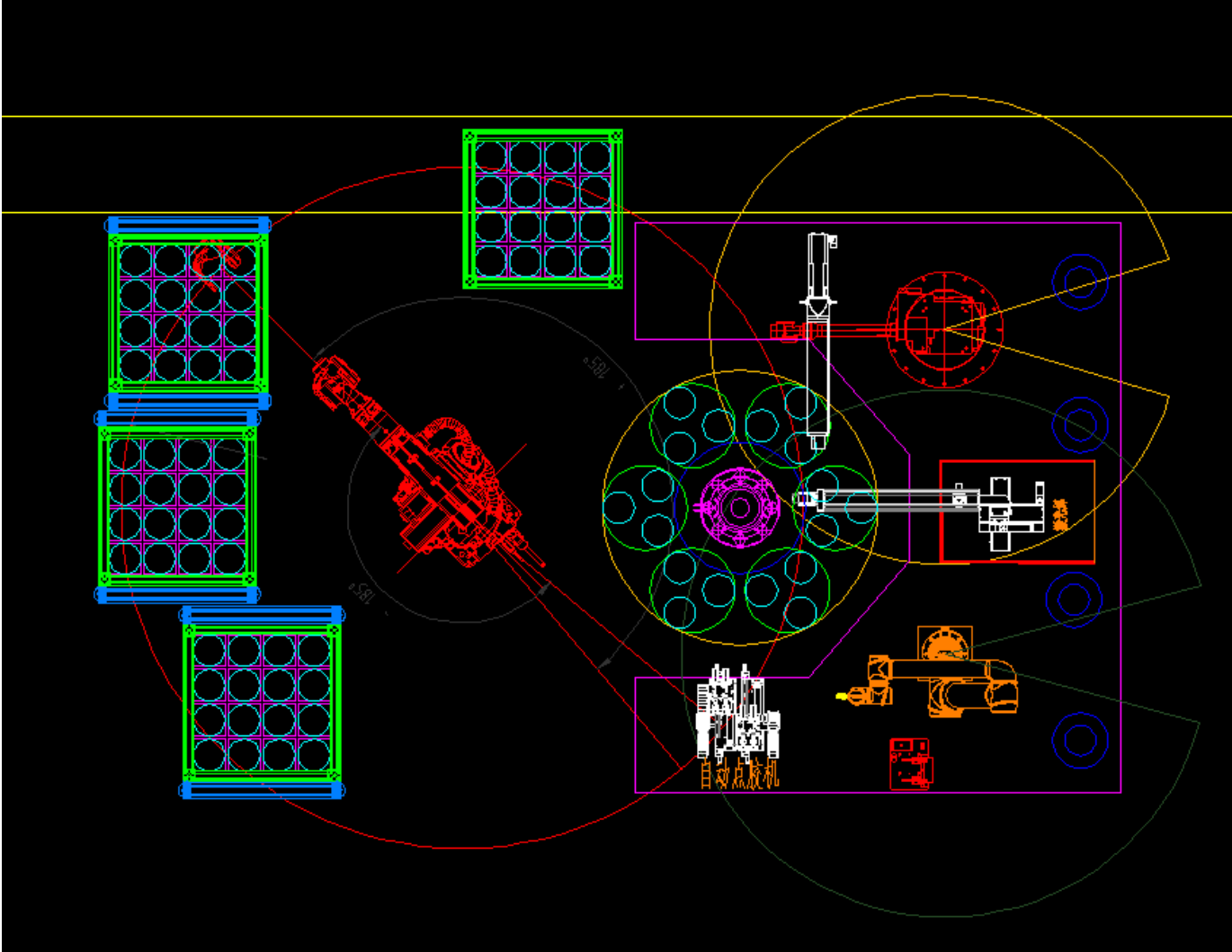


## 2. Workstation Layout Diagram





### 3. Floor Plan

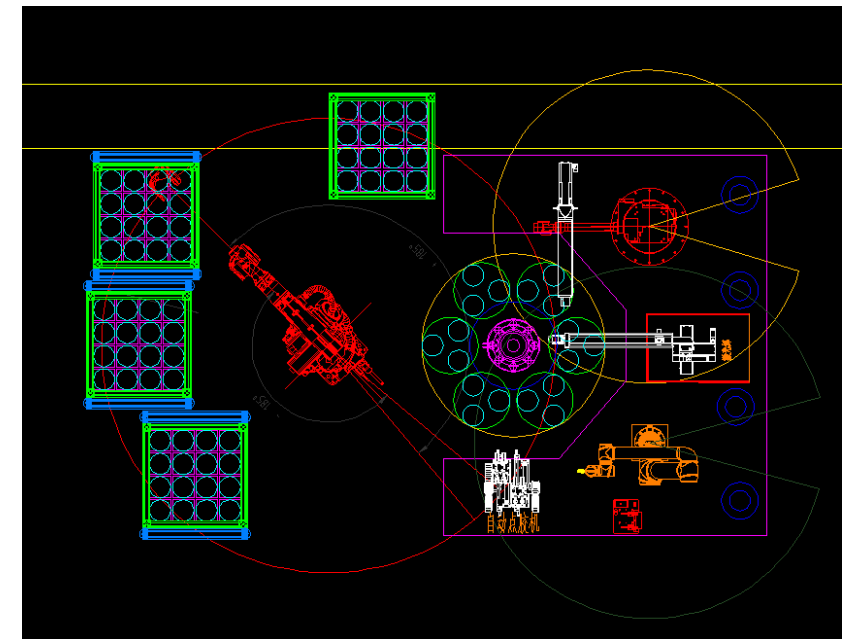
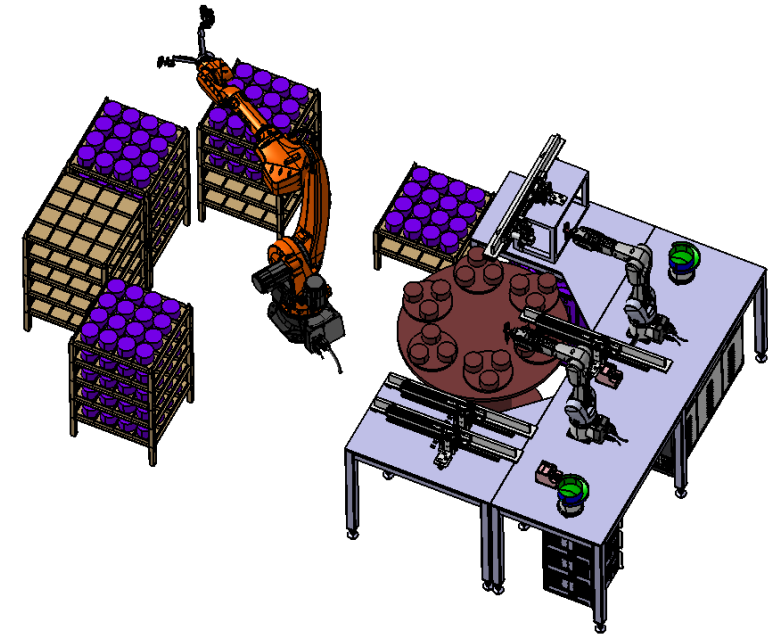




#### 4. Overall Solution Description

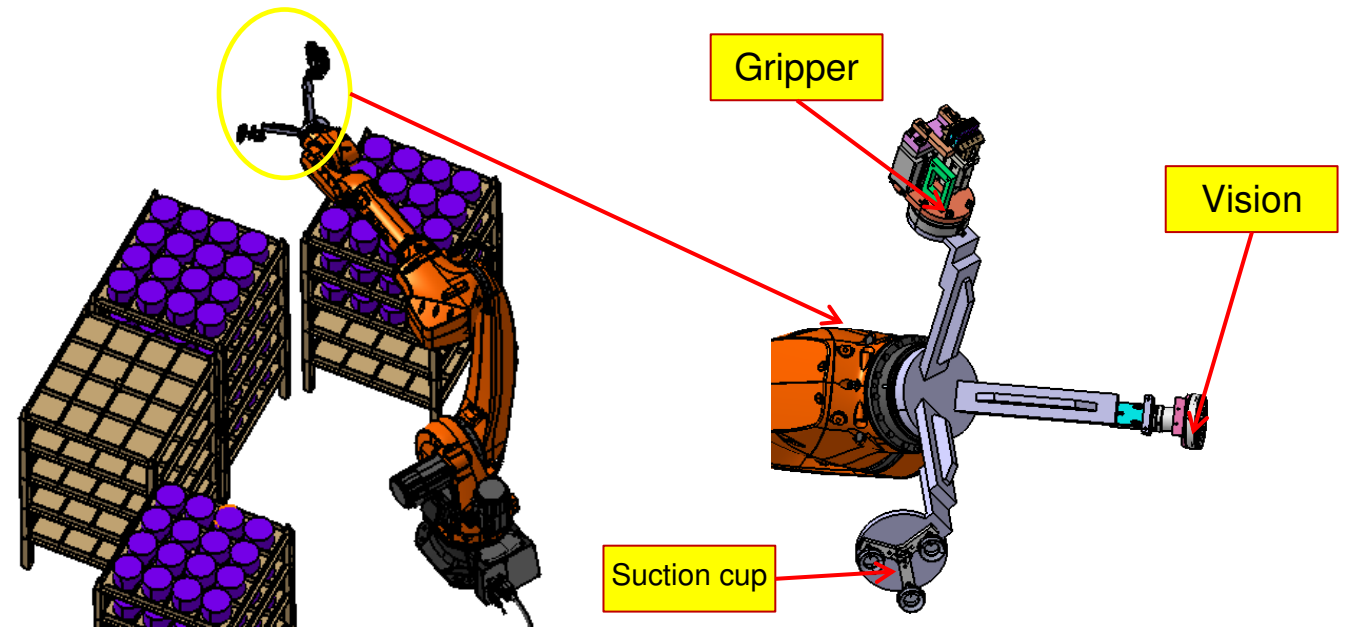
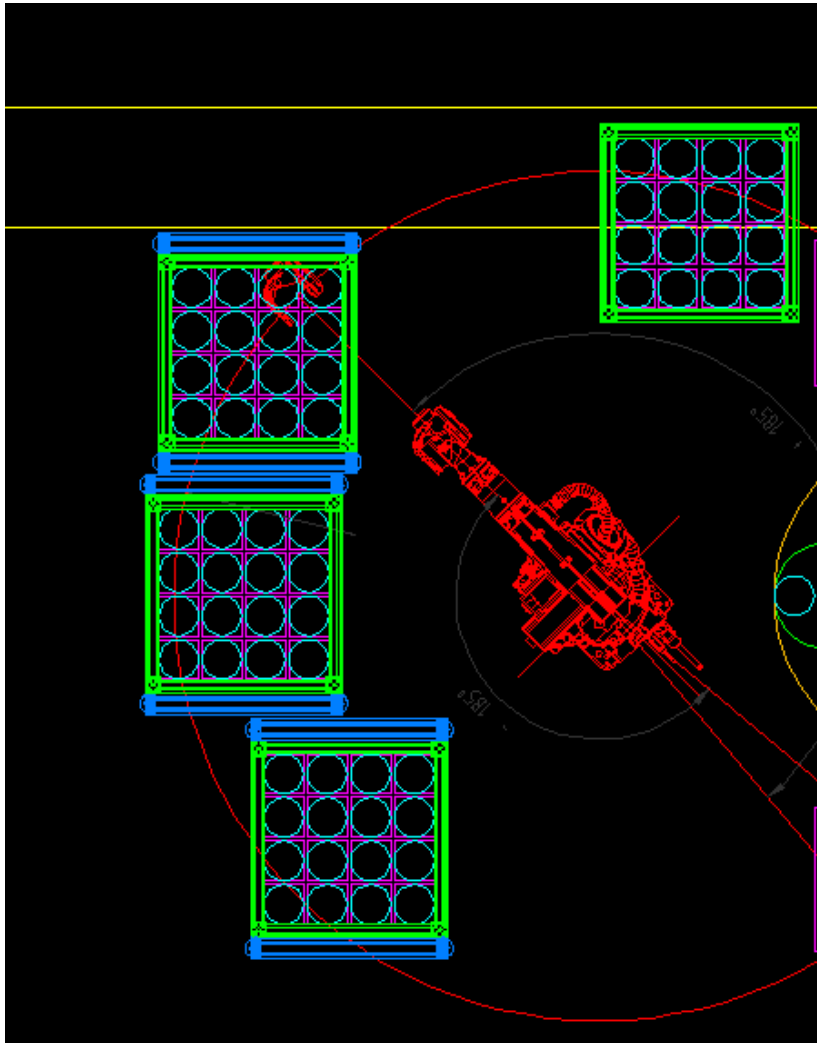
##### Solution Overview:

1. The solution follows a standardized, modular, flexible and automated design philosophy, maximally satisfying the customer's flexible-assembly needs across multiple products and parts;
2. The standardized workstation has 3 core areas;
  - Left — material in/out area: multiple material racks in a C-shaped arrangement with a handling robot in the middle; all racks are standardized and support changeover between many part types.
  - Center — core rotary table: the large mother plate divides into 6–8 index plates, each further into 3–4 small index plates, supporting 18–32 fixture changeovers; this flexible rotary plate enables loading/unloading of many part types and multi-step process flow.
  - Right — core process area: 1–2 four-axis process robots on a standard cabinet-type worktop perform tightening and other processes; a tool parking rack on the worktop holds swapped-out tools; robot, tool and vision controllers sit in the cabinet below, keeping the site tidy.





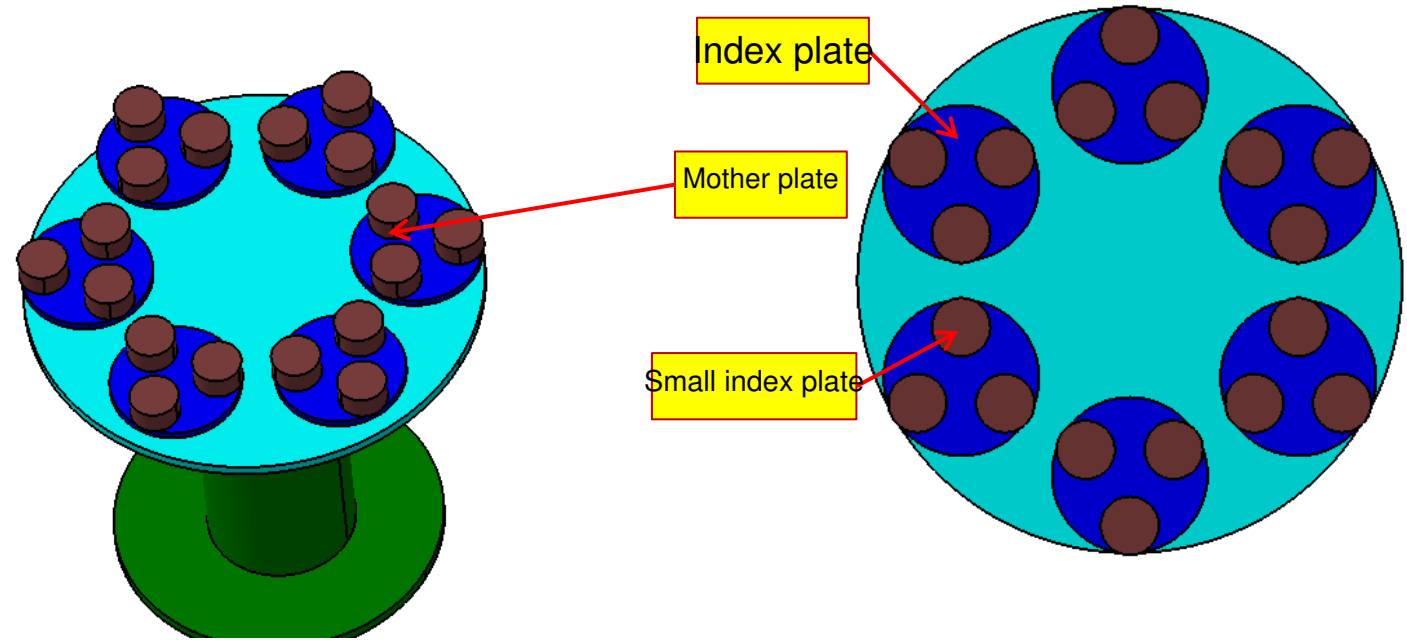
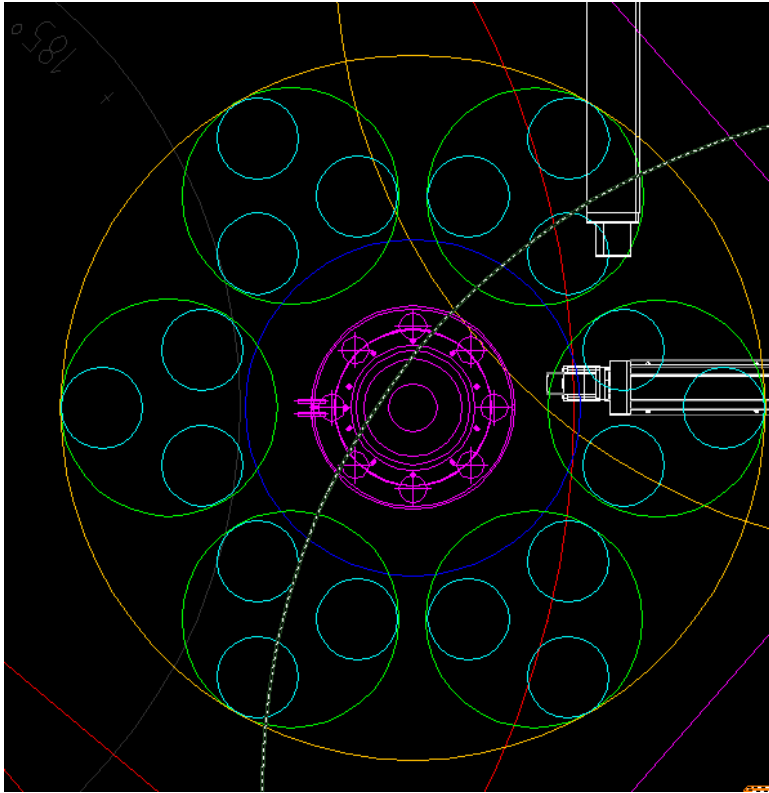
## 5. Detailed Description



- Material boxes use a C-shaped arrangement; multiple racks can be placed around the transfer robot. All racks are standardized, with stackers on both sides enabling upper/lower rack-level changeover.
- Racks support packaged line-feeding with standardized trays; one housing is paired with two small-part fixtures.
- The handling robot carries a combined tool — vision, gripper and suction cup mounted in three directions — for multi-functionality; tools can be quick-changed when many different parts must be handled.
- Vision guidance on the handling robot locates each part before picking, ensuring accurate grasping.



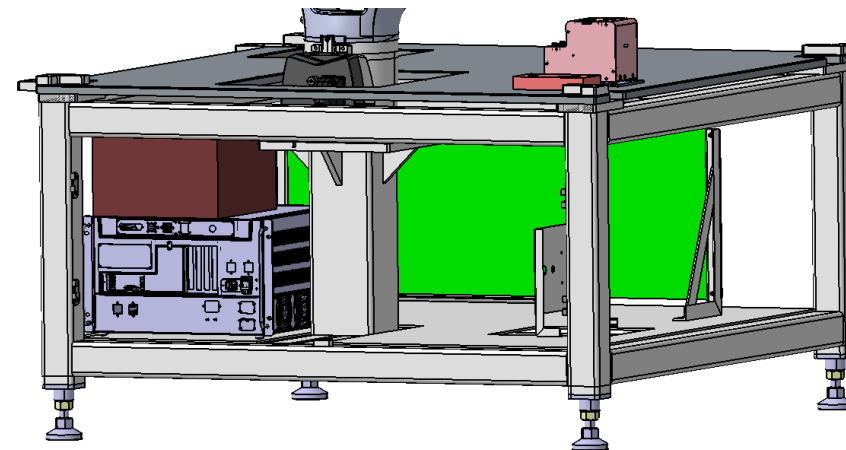
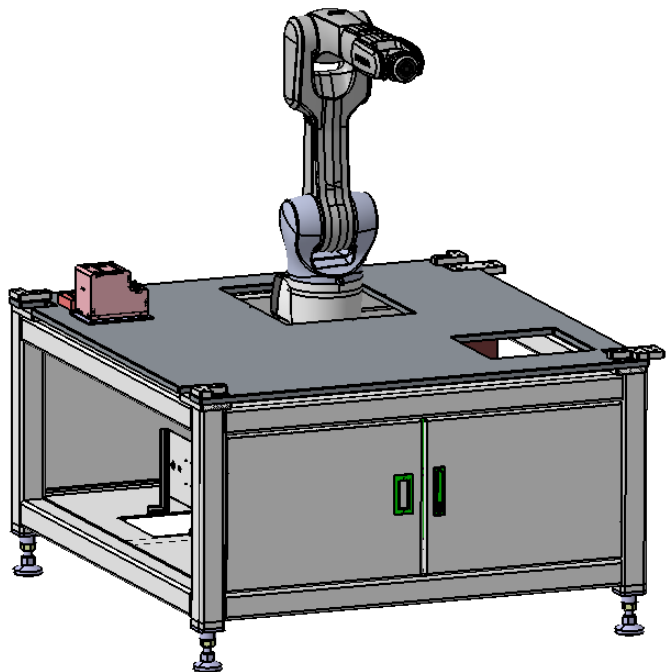
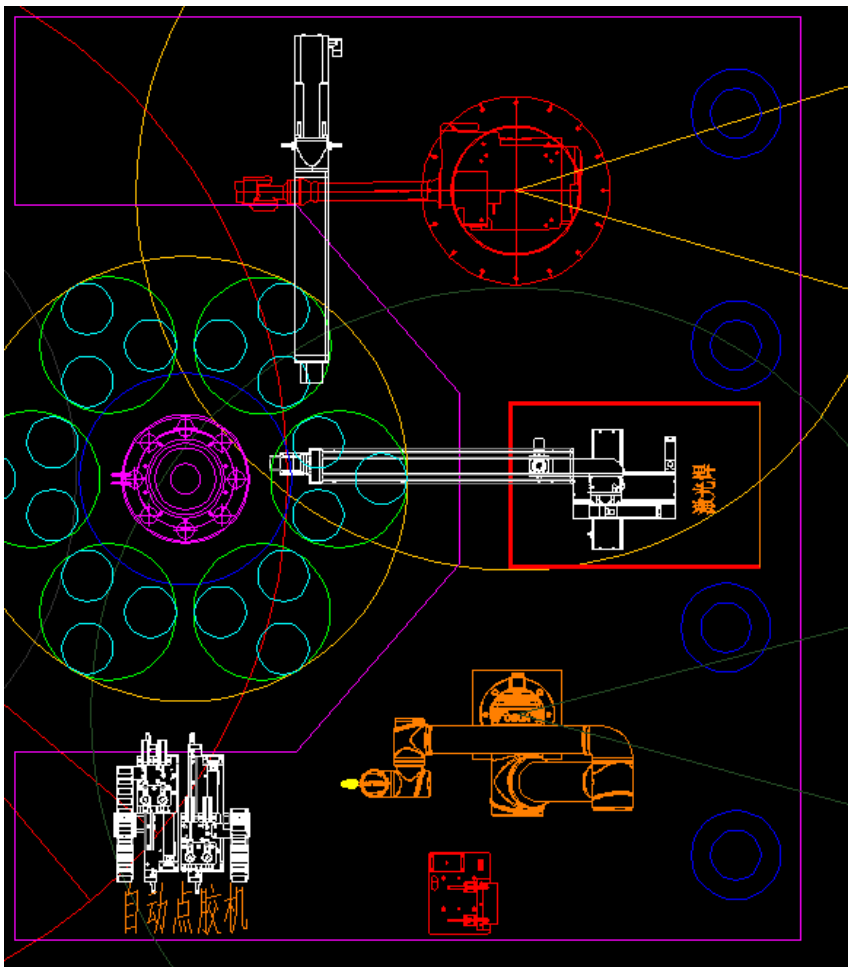
## 5. Detailed Description



- A large flexible mother plate of standardized design (diameter 1–1.3 m optional) is mounted on the rotary table. It divides into 6 index plates, each into 3 small index plates with standardized fixture interfaces, supporting 18 fixture changeovers; the flexible rotary plate enables loading/unloading of many part types and multi-step process flow.



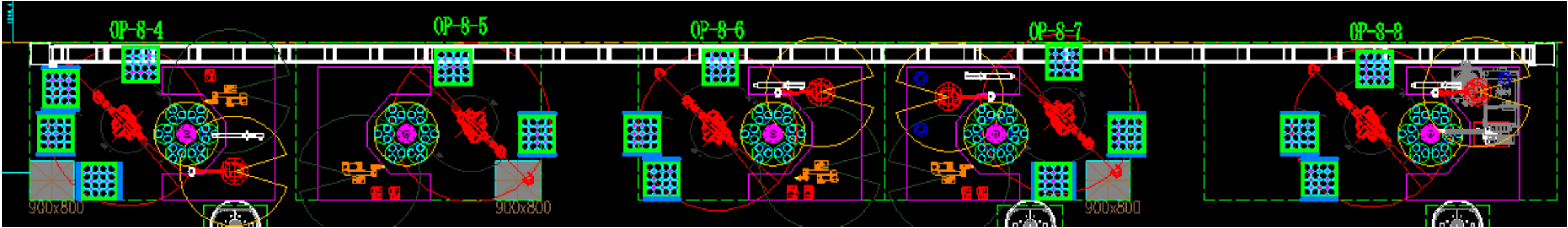
## 5. Detailed Description



- 1–2 four-axis process robots and other process equipment are placed on a standard cabinet-type worktop, enabling multi-purpose processing.
- A tool parking rack on the worktop holds swapped-out tools; robot, tool and vision controllers are housed in the cabinet below, keeping the site clean and tidy.



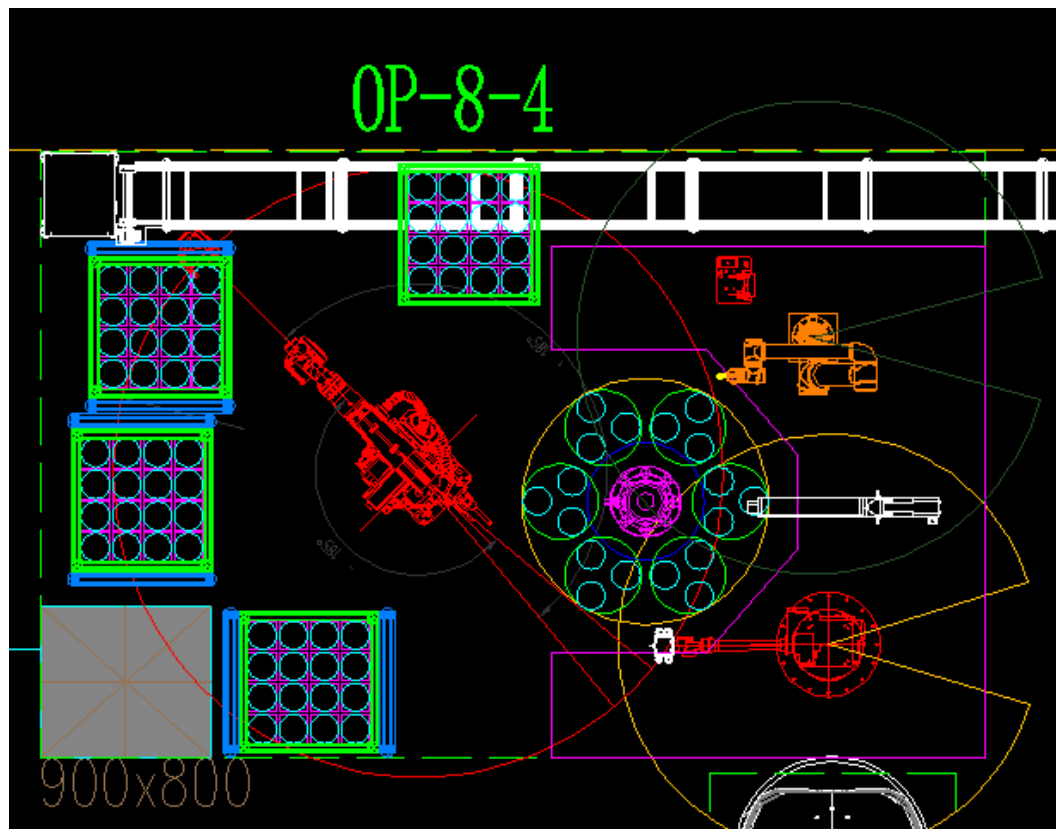
5. Line Layout Description





## 5. Line Layout Description

| Step  | Process description                                                                                 | Process equipment    | Equipment notes    | Material                                      | Size — actuator #8 | Size — #11 & #18 | Feeding module |
|-------|-----------------------------------------------------------------------------------------------------|----------------------|--------------------|-----------------------------------------------|--------------------|------------------|----------------|
| 8-4-1 | Pick drive gear 1 & motor-mounting-flange weldment, align to the motor screw holes and install      | Robot + vision       |                    | Motor                                         | D47-H17            | D68-H20.8        | Tray feeding   |
|       |                                                                                                     |                      |                    | Drive gear 1 & motor-mounting-flange weldment | D30-H8             | D36-H9.75        | Tray feeding   |
| 8-4-2 | Drive screws to fasten drive gear 1 & flange weldment onto the motor                                | Tightening equipment | 1 screw feeder set | Hex socket countersunk screw                  | M2×3               | M2×3             | Screw feeder   |
| 8-4-2 | Flip the motor; magnet is pre-bonded to the magnet mount; press-fit the magnet mount onto the motor | Press-fit equipment  | 2 press heads      | Magnet mount with pre-bonded magnet           | D12.4-H10.18       | D20-H24.55       | Tray feeding   |



[illegible]

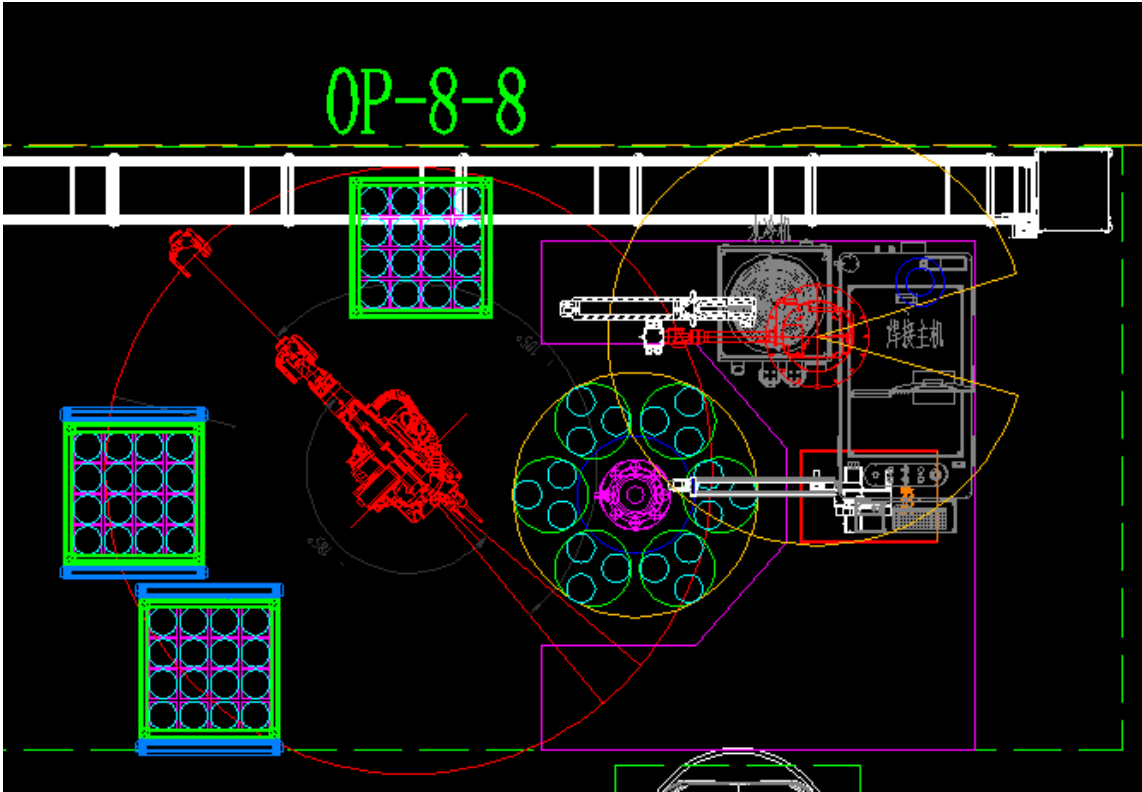






5. Line Layout Description

| Step  | Process description                                                                                                                                                                               | Process equipment     | Equipment notes | Material                                     | Size — actuator #8 | Size — #11 & #18 | Feeding module            |
|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|-----------------|----------------------------------------------|--------------------|------------------|---------------------------|
| 8-8-1 | Press-fit the drive-gear-3 & first-stage sun gear weldment onto the housing bearing; mesh the first-stage sun gear with the first-stage planetary gears, then mesh drive gear 3 with drive gear 2 | Robot + force control |                 | Drive gear 3 & first-stage sun gear weldment | D46-H14.75         | D58-H16.5        | Tray feeding              |
| 8-8-2 | Press-fit the bearing onto drive gear 3                                                                                                                                                           | Press-fit equipment   | 1 press head    | Bearing                                      | D23-H4             | D23-H4           | Tray feeding              |
| 8-8-3 | Press-fit the end drive gear onto the bearing of drive gear 3                                                                                                                                     |                       |                 | End drive gear                               | D20-H5.5           | D21-H5.75        | Flexible vibratory feeder |
| 8-8-4 | Weld the end drive gear                                                                                                                                                                           | Laser welding         |                 | /                                            | /                  | /                | /                         |





## 6. Software Standard SICAR



Siemens SICAR standardization differs from ECAD: the automation standard covers not only software but also hardware, network architecture, process, program libraries, HMI templates and more.

End users:

- Lower downstream personnel-training and equipment-maintenance costs
- Consistent, higher quality across automation equipment and lines
- Predefined hardware and software shorten engineering time and improve efficiency

OEMs:

- Pre-installed cabinets and function units shorten delivery time (MOP & Drivebox)
- Standardized installation rules and documentation let OEMs finish installation faster with less repetitive work
- Long-term customer–supplier relationships

### Current automation standard applications



Mercedes-Benz



**SIEMENS**



BMW L Standard





## 6. Software Standard SICAR

# SICAR@TIA Portal

## Overview



### Automation lines



### Conveying

Car plants + OEMs  
without their own automation standard → based on TIA Portal



## SICAR

### HW/SW platform



Step 7 V13/V14



S7-1513F~1518F



WinCC Adv. V13/V14



SINAMICS

Base program  
Executable base  
framework project  
PLC + HMI = Start



Scalability

### Graph LAD FBD

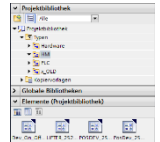


Graph  
LAD  
FBD

Simple architecture

### Library

- PLC templates
- HMI panels



### Integrated system functions

- System diagnostics
- Code views (LAD/ FBD/GRAPH)
- Integrated message system



### Docs

- Online help (Step7 / WinCC RT)

• ...





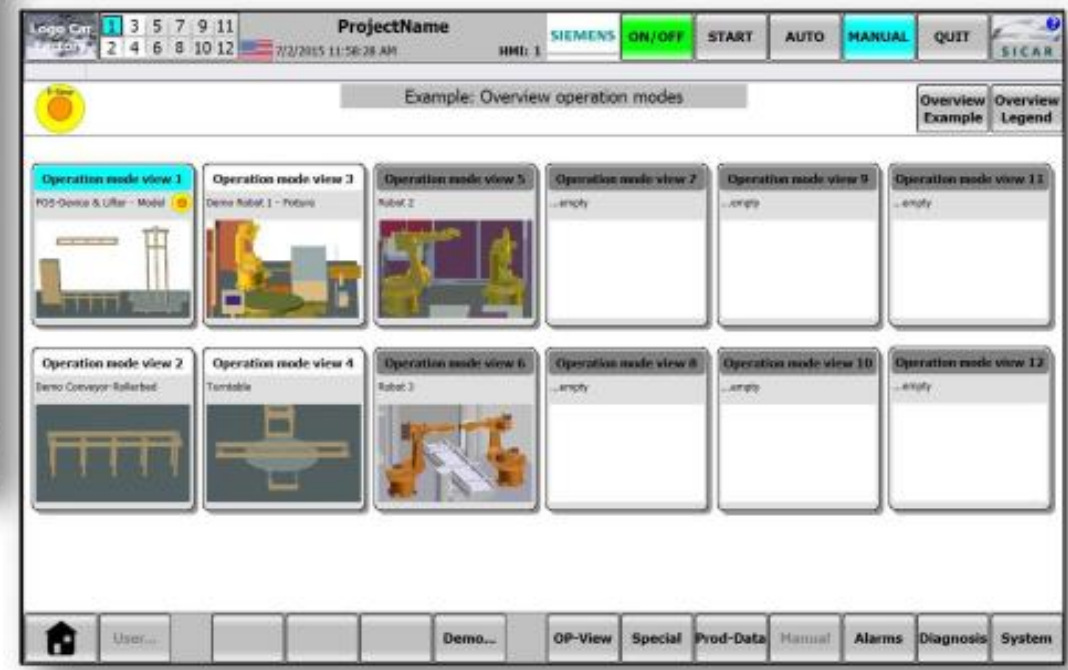
## 6. Software Standard SICAR

- Visualization UI — station overview / selection:

样例: 工厂概览



样例: 操作模式概览 (动态的)





SICAR is widely used across the automation industry; most Chinese own-brand and NEV automakers use Siemens SICAR as the automation standard in their plants. For end owners and integrators, standardization brings the following benefits:

- Line builders:
- A unified programming approach across projects improves efficiency and saves training cost
- Standardized programs and documentation let OEMs commission equipment faster with less repetitive work
- Long-term customer–supplier relationships

End users:

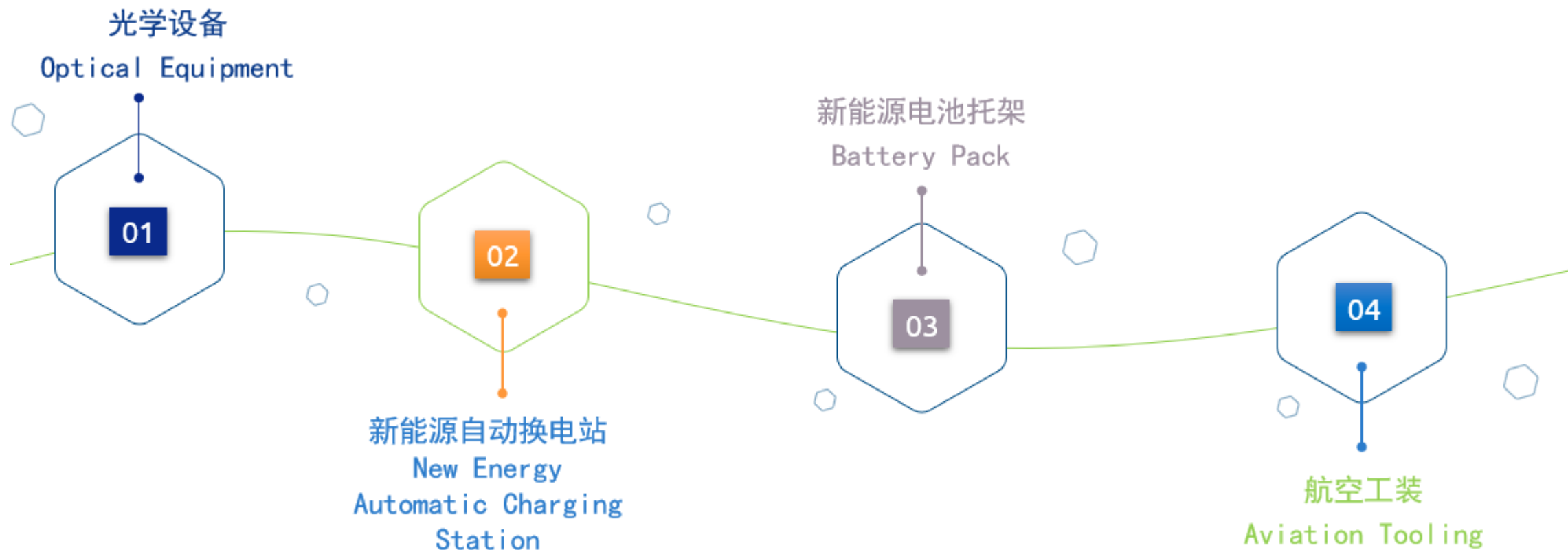
- Lower downstream personnel-training and equipment-maintenance costs
- Consistent, higher quality across automation equipment and lines

- Predefined hardware and software shorten engineering time and improve efficiency

- Standardized programs on cost-effective Siemens CPUs reduce cost.
- Product changeover needs no development — recipes enable fast changeover.
- Highly scalable — one station can adapt to dozens of fixture types.
- Low entry barrier — process programs can be configured without programming experience.

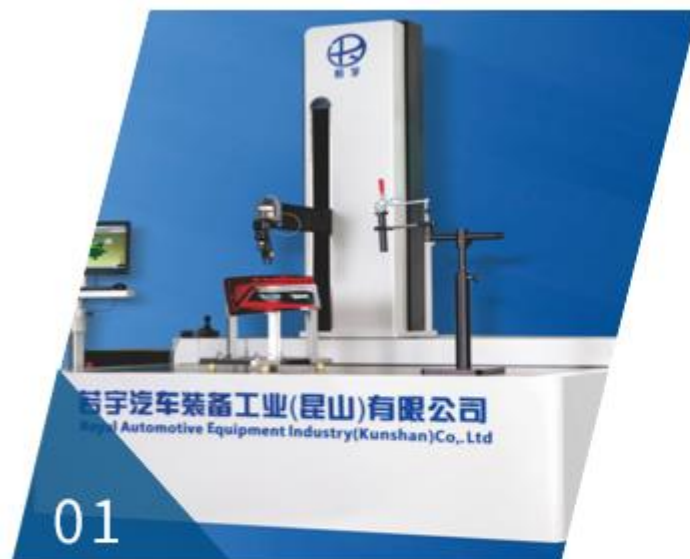


## 7. Ruoyu Product & Project Cases





## Optical Equipment



透明材质光学检测仪  
Optical Detector for  
Transparent Material



手持式无线3D光学扫描仪  
Handheld Wireless 3D  
Optical Scanner



## New Energy Automatic Battery-Swap Station



新能源换电站  
New Energy Charging Station



新能源换电站  
New Energy Charging Station

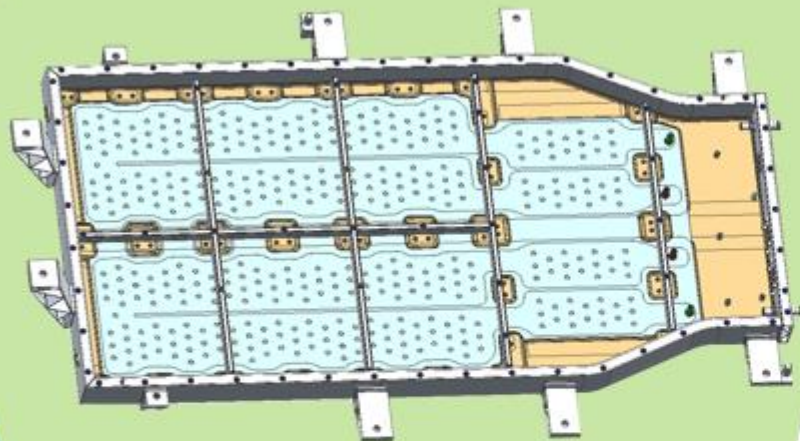


新能源换电站  
New Energy Charging Station



## Battery Pack Tray & Aviation Tooling

01



新能源车托架  
Battery Pack

02

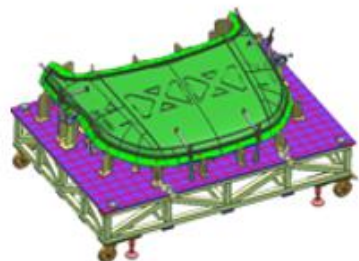


航空工装  
Aviation Tooling

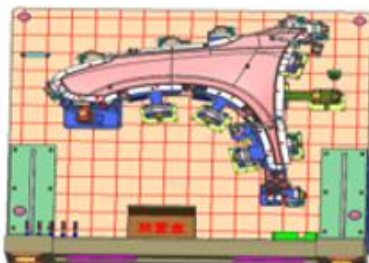


## 7. Ruoyu Product & Project Cases

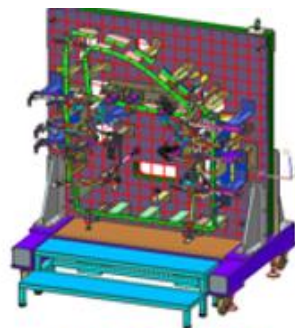
Precision Manufacturing — Precision Locating Fixtures



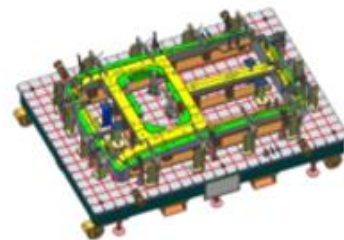
● 前盖总成检具  
Front Cover  
Assy Gauge



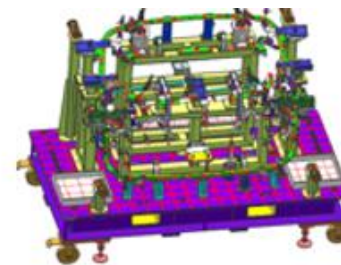
● 翼子板总成检具  
Fender Assy  
Gauge



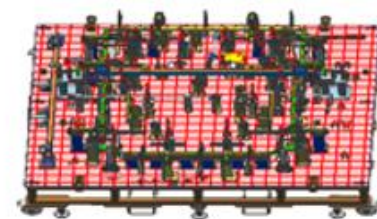
● 前门总成检具  
Front Door  
Assy Gauge



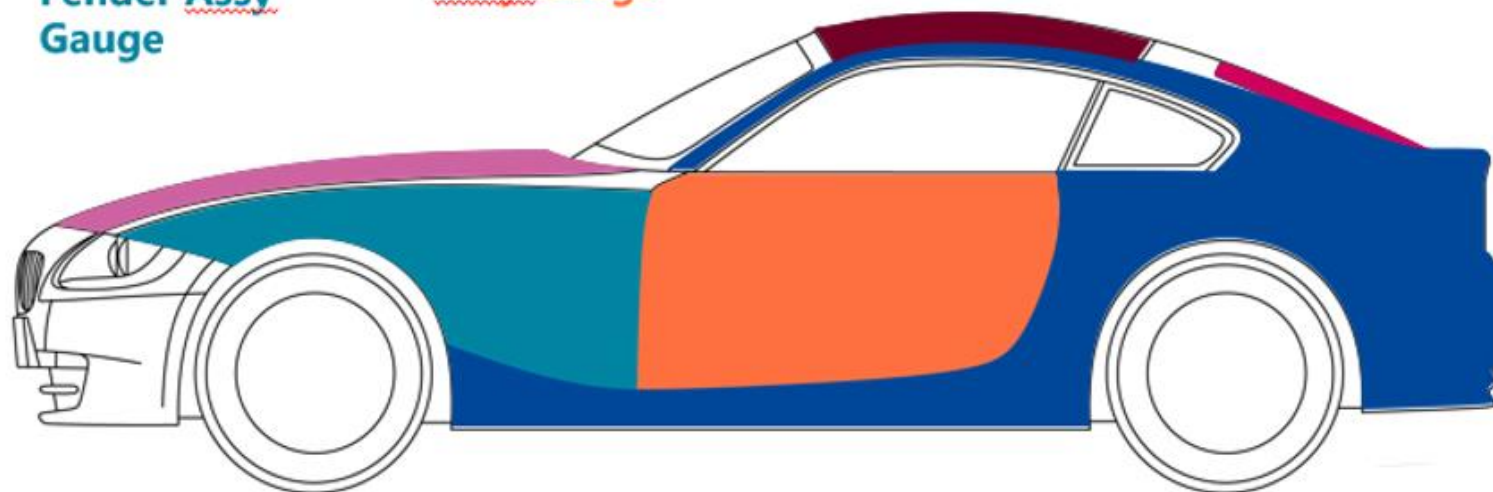
● 天窗总成检具  
Sunroof Assy  
Gauge



● 后盖总成检具  
End Cover  
Assy Gauge



● 后地板总成  
Rear Floor  
Assy Gauge





## 7. Ruoyu Product & Project Cases

### Automation — Hemming Machine



Hemming  
machine



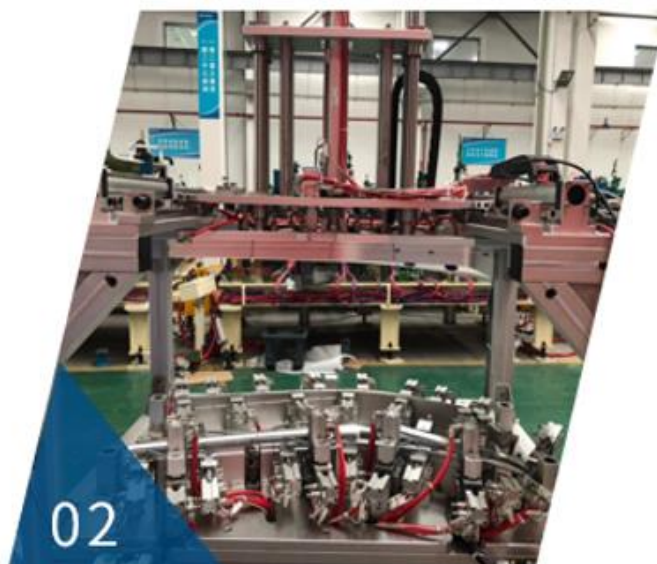
Hemming  
machine



## Automation — Line Automation



自动化工装  
Automatic Tooling

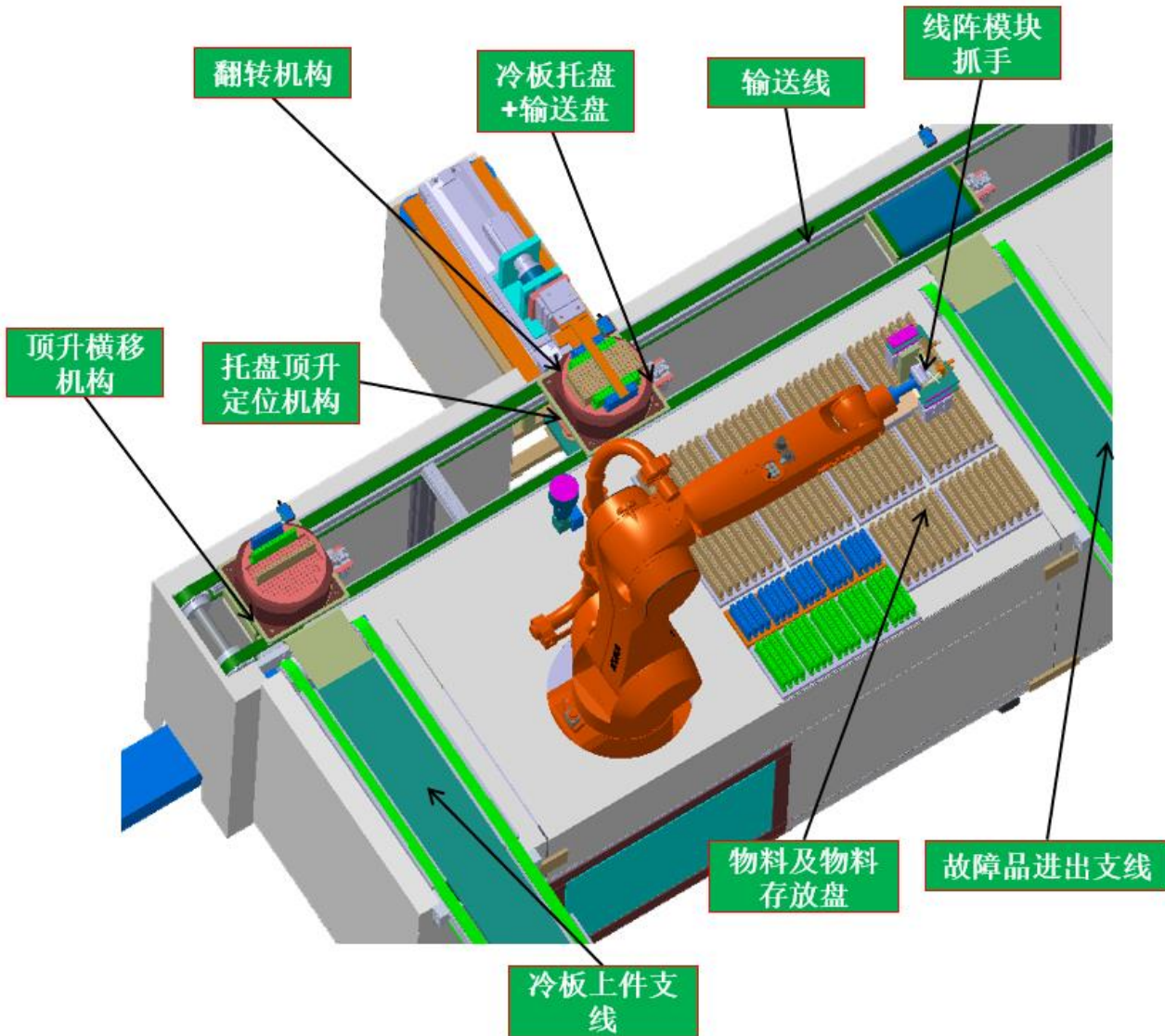


自动化工装  
Automatic Tooling



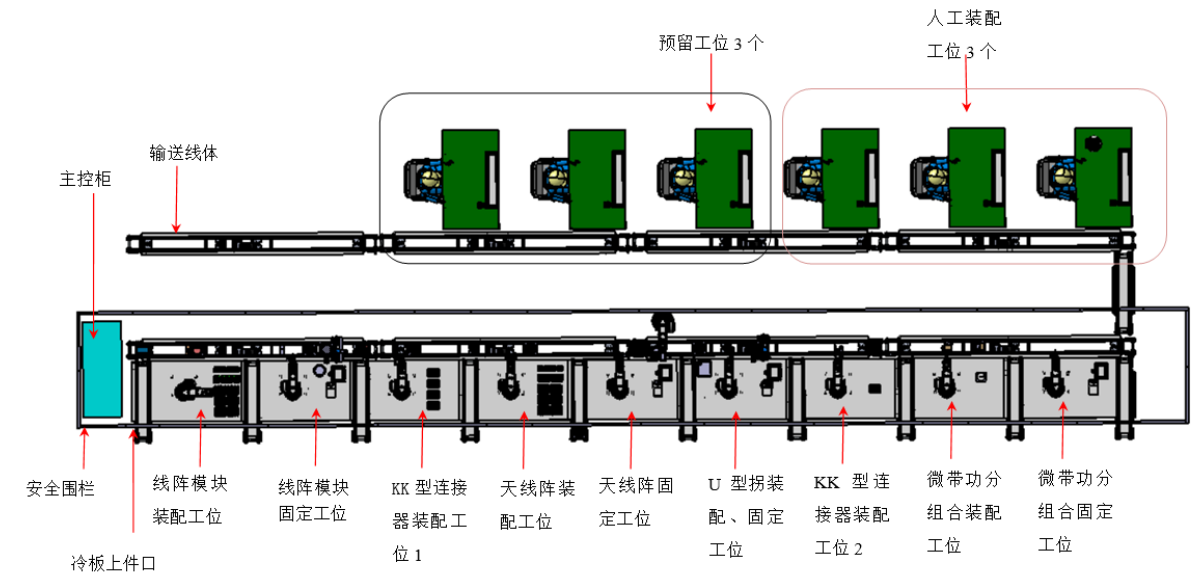


7. Ruoyu Product & Project Cases



Automation — Aerospace & Defense

Phased-array antenna automated assembly line





# 制造能力

## Manufacturing Capacity



### 加工设备Processing Equipment

最大五轴机床尺寸:

Size of the largest 5-axis machine tool:  
4060\*3200\*1200

最大三轴机床尺寸

Size of the largest 3-axis machine tool :  
6000\*2500\*800

拥有各类加工设备150多台, 五轴五台



### 制造人员Manufacturing Staff

员工共240余人。More than 240 staff totally

CAM编程工程师、操作人员59人。CAM programmers and operators: 59

工艺工程师8人。Process engineers: 8

钳工装配7大组, 资深经验钳工36人。

Seven groups for benchwork and assembly. Rich-experience fitters: 36  
月制造能力可达300多套。

Monthly manufacturing capacity can reach over 300 sets.





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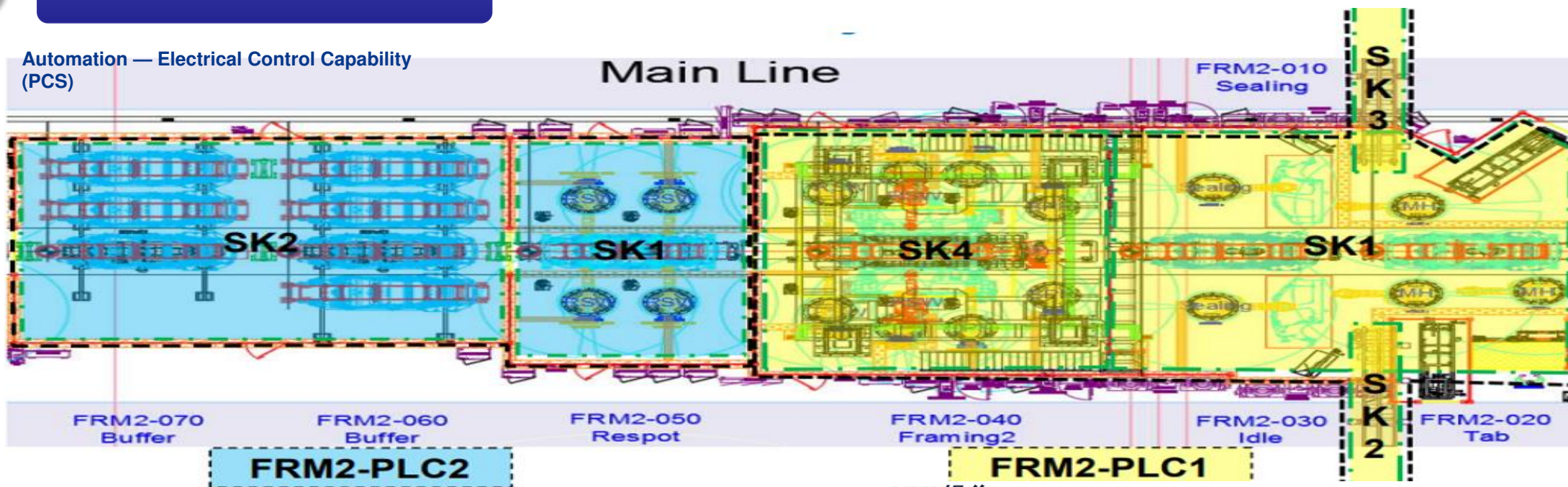
Monthly manufacturing capacity can reach over 300 sets.





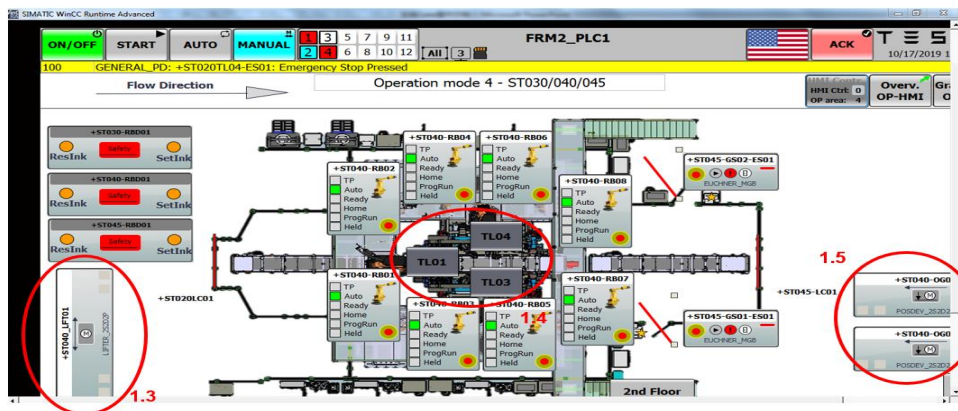
## 7. Ruoyu Product & Project Cases

Automation — Electrical Control Capability  
(PCS)



HMI操作

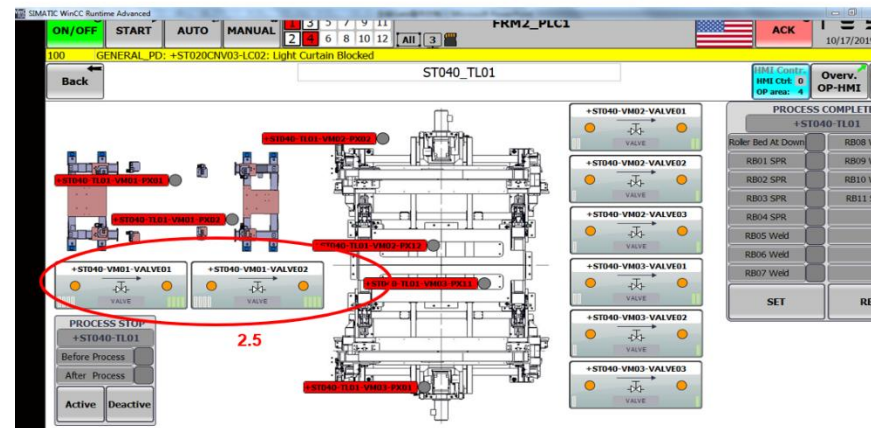
1.HMI手动模式操作Gate（合拢）



S4：确认Gate区域的初始状态

HMI操作

2.HMI手动模式操作



S6：点击2.2，进入TL01，点击2.5，操作打开两组气缸

Electrical-control PCS scope of three automated main-line areas in the Tesla Shanghai Gigafactory shop, covering hardware design, software-standard development, and on-site installation & commissioning services.



# 测量能力

## Manufacturing Capacity



### 检测设备Testing Equipment

三坐标测量机CMM:

龙门测量机尺寸/Size of gantry CMM:

5000\*2500\*1800

悬臂测量机尺寸/Size of cantilever CMM:

6000\*3000\*2500

测量设备达40臂

Measuring Equipment:40 arms



### 检测人员Testing Staff

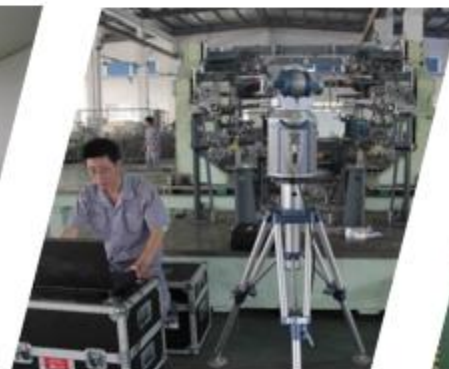
品控部有工程师90名，其中测量工程师45名，

调试工程师30名，其他15名。公司月测量可达250多套。

There are 90 engineers in QC department, including 45

measuring engineers, 30 commissioning engineers and 15

others. Monthly measurement capacity can reach more than 250 sets.



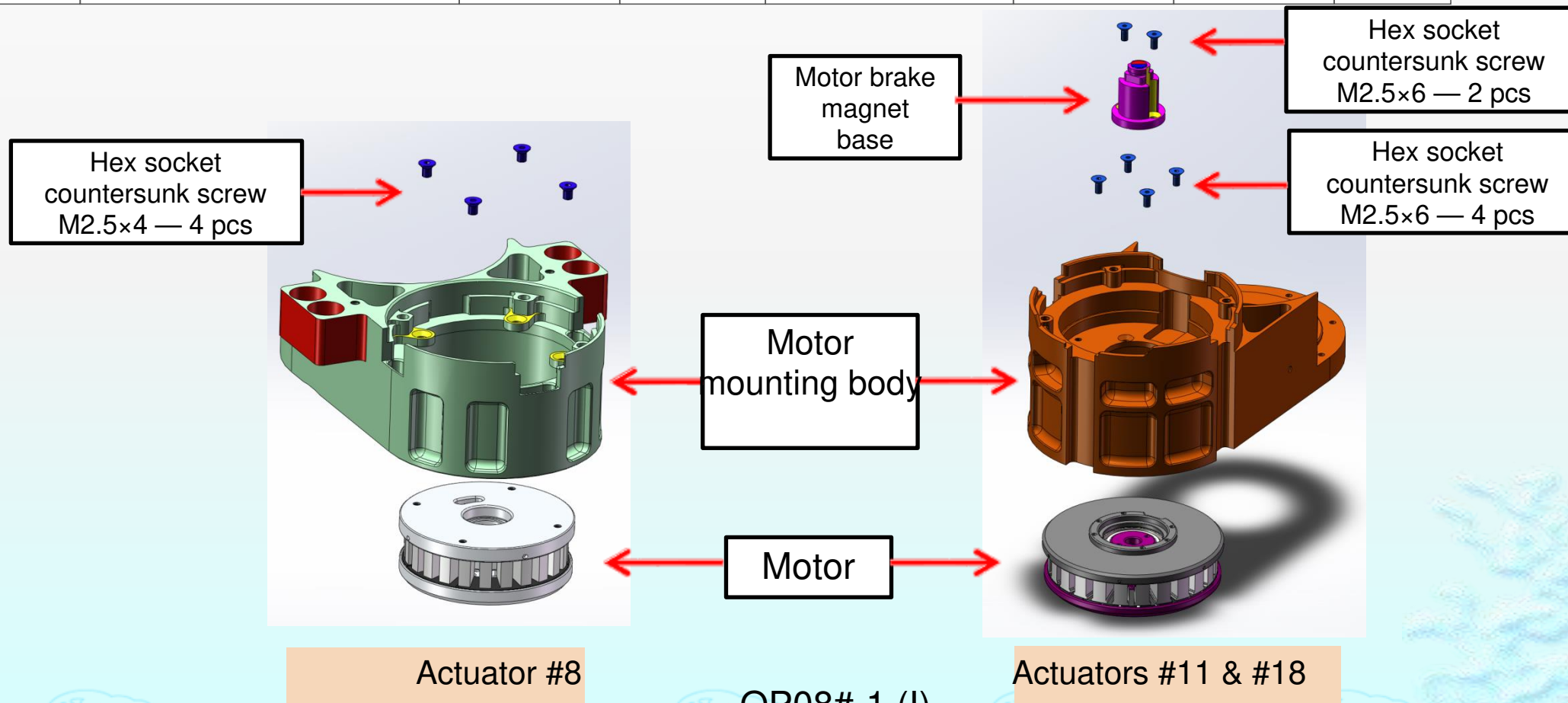
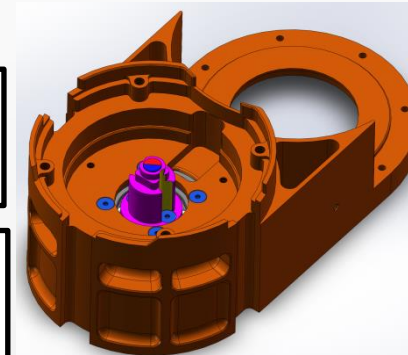
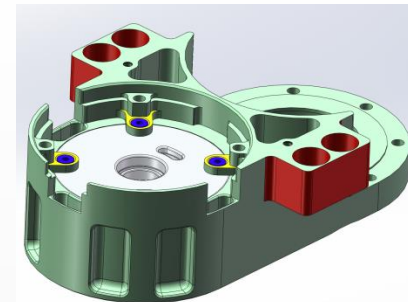


**Thank you for your attention!**



## OP08#-1 Product Description

| Step  | Process description                                                                            | Process equipment                        | Equipment notes     | Material                     | Material size             |                     | Feeding module |
|-------|------------------------------------------------------------------------------------------------|------------------------------------------|---------------------|------------------------------|---------------------------|---------------------|----------------|
|       |                                                                                                |                                          |                     |                              | Actuator #8               | Actuators #11 & #18 |                |
| 8-1-1 | Pick the motor onto the fixture; straighten the motor cables                                   | Robot + vision, cable-dressing mechanism |                     | Motor mounting body          | L120-W86.8-H33.75         | L155-W80-H53.6      | Tray feeding   |
| 8-1-2 | Pick the motor housing, sleeve it over the motor, align the screw holes and tighten (4 screws) | Robot + vision, tightening equipment     | 2 screw feeder sets | Motor                        | D47-H17                   | D68-H20.8           | Tray feeding   |
|       |                                                                                                |                                          |                     | Hex socket countersunk screw | M3×4 (changed to: M2.5×4) | M2.5×6              | Screw feeder   |
| 8-1-3 | Install the motor brake magnet base on the motor and tighten (2 screws)                        | Tightening equipment                     | 1 screw feeder set  | Motor brake magnet base      | /                         | D20-H24.55          | Tray feeding   |
|       |                                                                                                |                                          |                     | Hex socket countersunk screw | /                         | M2.5×6              | Screw feeder   |





## OP08#-1 Process Flow & Sequence Overview

(Turntable rotates 120°)

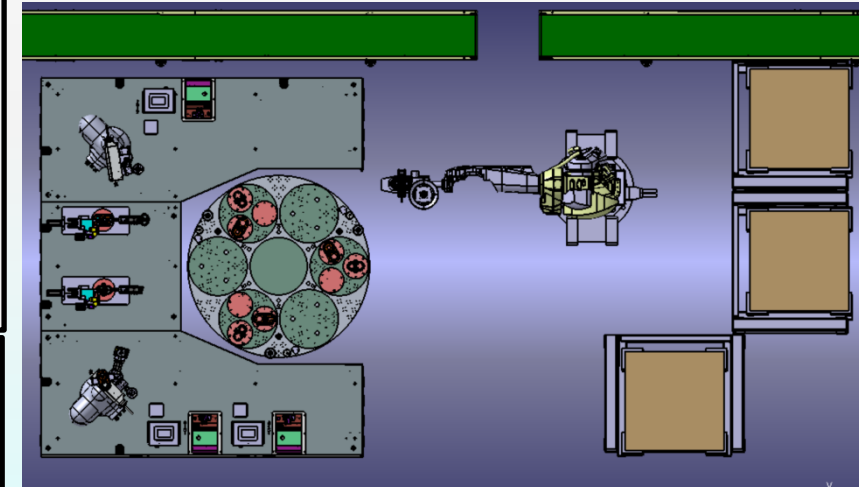
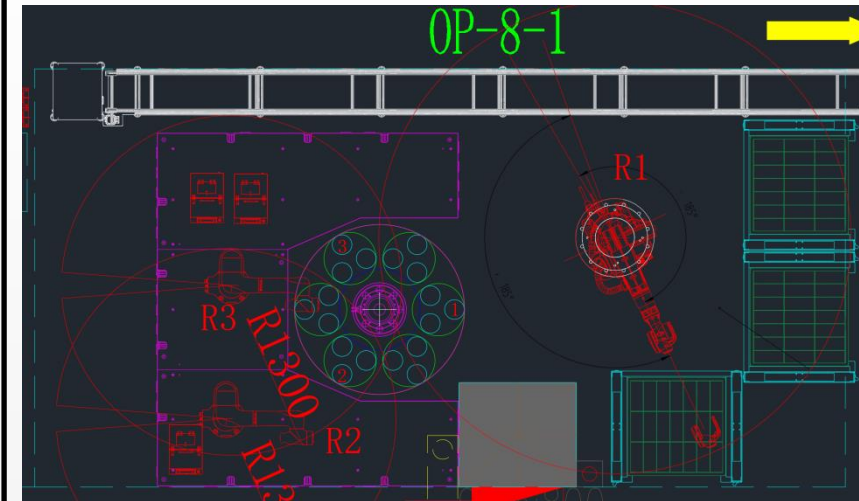
1. Robot R1 takes the assembled product from the turntable fixture and places it on the conveyor.
2. R1, vision-guided, takes a motor mounting body from the stacker tray onto the turntable fixture;
3. R1, vision-guided, takes a motor from the stacker tray onto the turntable transfer fixture;
4. R1, vision-guided, takes a motor brake magnet base from the stacker tray onto the turntable transfer fixture (only for actuators #11 & #18); (turntable rotates 120°)
5. Robot R2 takes the motor from the turntable transfer fixture onto the workbench fixture;
6. The auxiliary cable-dressing fixture arranges the motor cables;
7. R2 takes the motor mounting body from the turntable fixture and assembles it onto the motor in the workbench fixture (as the body descends to height, the cable-dressing fixture retracts from the interference zone), aligning the locking screw holes per assembly orientation;
8. The auxiliary clamping fixture clamps the motor / motor mounting body;
9. R2 picks hex socket countersunk screws M2.5×4 from the screw feeder (M2.5×6 for actuators #11 & #18) (×4);
10. R2 automatically tightens the motor mounting body to the motor (×4);
11. The clamping fixture releases the assembled product;
17. R2 returns the assembled product from the workbench fixture to the turntable fixture; (turntable rotates 120°)
12. Robot R3 takes the motor brake magnet base from the turntable transfer fixture onto the workbench fixture (aligned to the motor shaft hole), aligning screw holes per assembly orientation; (only for #11 & #18)
13. The clamping fixture clamps the brake magnet base on the product; (only for #11 & #18)
14. R3 picks hex socket countersunk screws M2.5×6 from the screw feeder (×2); (only for #11 & #18)
15. R3 automatically tightens the brake magnet base onto the motor (×2); (only for #11 & #18)
16. The clamping fixture releases the product (with brake magnet base installed); (only for #11 & #18)

### Notes summary:

R1: FANUC robot; R2: ROKAE robot; R3: ROKAE robot.

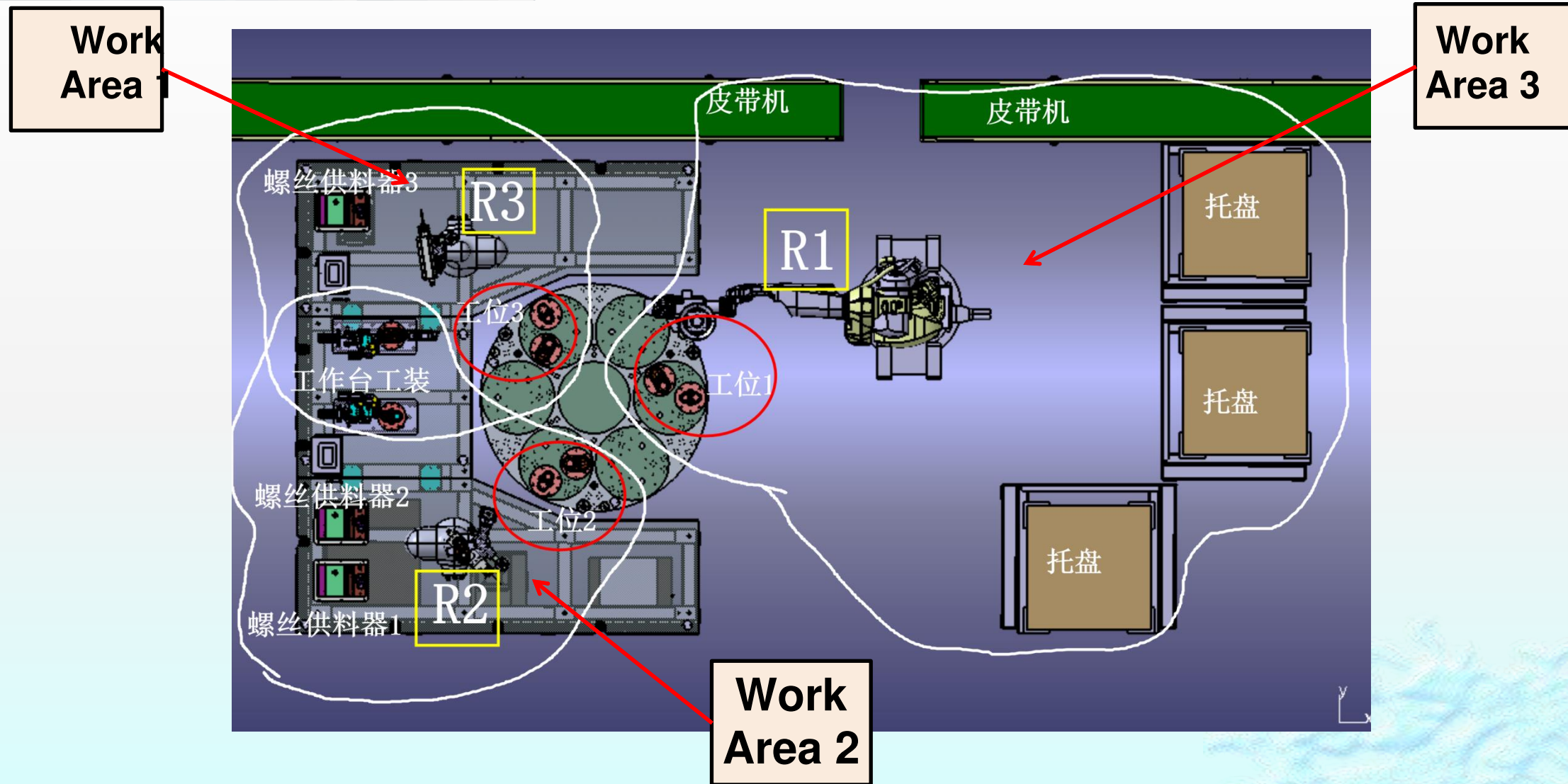
① R2 handles installation and tightening of the motor and brake magnet base with collaborative assistance;  
R3 handles installation of the motor mounting body and transfers the finished product from workbench to turntable;

②





## 1. Station and work area division





## 2. Station equipment & work area description

### Work Area 1: robot R1 operating area

Within this area, per the station cycle-time analysis table, R1:

- A. Unloads finished product assemblies from Station 1 to the belt conveyor;
- B. Loads parts from trays onto the Station 1 fixture;
- C. Assists the stacker with empty/full (or full/empty) tray changeover.

### Work Area 2: robot R2 operating area

Within this area, per the station cycle-time analysis table, R2:

- A. Picks parts from the Station 2 fixture onto the workbench fixture and performs the specified assembly tasks;
- B. Picks screws from the screw feeder and tightens the product on the Station 2 fixture;
- C. Transfers the assembled product from the workbench fixture to the turntable Station 3 fixture.

### Work Area 3: robot R3 operating area

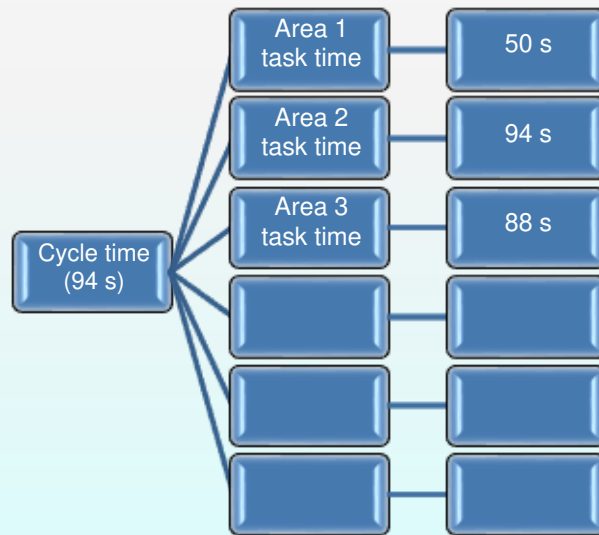
Within this area, per the station cycle-time analysis table, R3:

- A. Picks the brake-magnet mounting base from the Station 3 fixture onto the workbench fixture and performs the specified assembly tasks;
- B. Picks screws from the screw feeder and tightens the product on the workbench fixture.
- C.

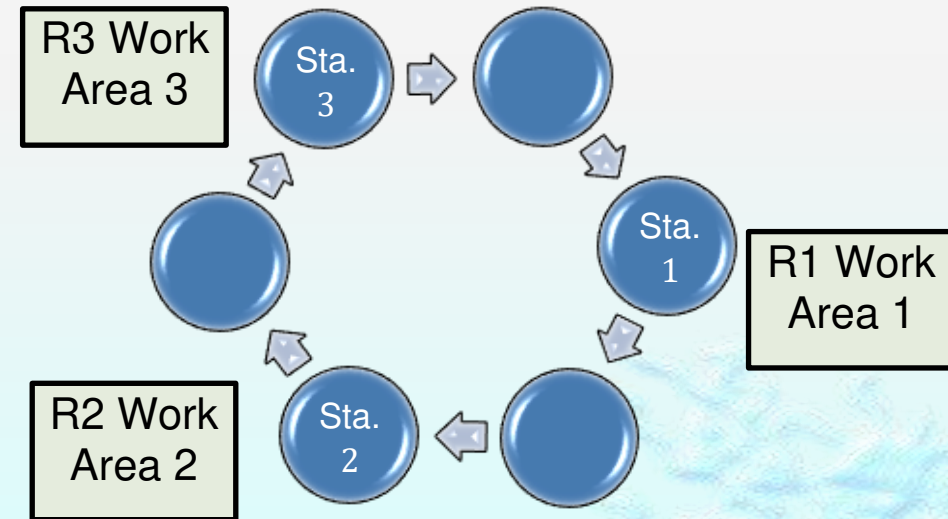


### 3. Station work logic

0. On power-up, every device self-checks and homes (HOME / origin position), then waits for dispatch commands;
1. The large turntable rotates 120° in the station's logical direction, stops, reports ready, and waits for dispatch;
2. Within one cycle period, each device executes its tasks in process order at its designated station and work area per the cycle-time table;
3. When the cycle ends and all devices have completed their tasks and reported ready, the dispatch system commands the next production cycle.
4. Steps 1–3 repeat in a loop.



In-cycle task duration plan



Process logic sequence



## 1. Station equipment list

Robot Assembly Standard Workstation — Equipment List

| Equipment | Handling robot | Handling vision | Flexible rotary table | Workbench | Box stacker | Trays | Flexible vibratory feeder | Vibratory feeder vision | Small process robot (6-axis, ROKAE) | Cobot (AUBO) | Assembly vision | Gear-meshing robot (TM) | Press | Laser welder | Screw feeders | Tightening equipment | Tightening vision | Bit changer |
|-----------|----------------|-----------------|-----------------------|-----------|-------------|-------|---------------------------|-------------------------|-------------------------------------|--------------|-----------------|-------------------------|-------|--------------|---------------|----------------------|-------------------|-------------|
| OP-8-1    | 1              | 1               | 1                     | 1         | 3           | 15    | 0                         | 0                       | 2                                   | 0            | 0               | 0                       | 0     | 0            | 3             | 2                    | 2                 | 1           |

## 2. Commissioning material list

| Level | Part No.        | Step  | Material                     | Material size               | Qty / set | Commissioning qty (×30) | Remarks                                             | Actuator  |
|-------|-----------------|-------|------------------------------|-----------------------------|-----------|-------------------------|-----------------------------------------------------|-----------|
| 2     | NM2000000300023 | 8-1-1 | Motor mounting body          | L120-W86.8-H33.75           | 1         | 30                      |                                                     | #8        |
| 2     | NM2000000300086 | 8-1-1 | Motor mounting body          | L155-W80-H53.6              | 1         | 30                      |                                                     | #11 & #18 |
| 2     | NA3000000000042 | 8-1-2 | Motor                        | D47-H17                     | 1         | 30                      | (40 amorphous motor)                                | #8        |
| 2     | NA3000000000043 | 8-1-2 | Motor                        | D68-H20.8                   | 1         | 30                      | (60 amorphous axial motor)                          | #11 & #18 |
| 2     | SCRW00000000011 | 8-1-2 | Hex socket countersunk screw | M2.5×4 (process card: M3×4) | 4         | 120                     | (non-standard head diameter) (screw head dia. Φ4.6) | #8        |
| 2     | SCRW00000000028 | 8-1-2 | Hex socket countersunk screw | M2.5×6                      | 4         | 120                     | (screw head dia. Φ5.6)                              | #11 & #18 |
| 2     | NM2000000300082 | 8-1-3 | Motor brake magnet base      | D20-H24.55                  | 1         | 30                      |                                                     | #11 & #18 |
| 2     | SCRW00000000028 | 8-1-3 | Hex socket countersunk screw | M2.5×6                      | 2         | 60                      | (screw head dia. Φ5.6)                              | #11 & #18 |